



Airbus Pléiades Neo Basic MS ORT: Quality Processes Review

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1 Introduction

1.1 Scope

This document summarises a Data Product Quality Processes Review for Airbus Pléiades Neo as part of the EODH Data Product Quality Process [AD-1]. The purpose is to demonstrate that the product has been developed and generated following processes that adhere to best practices.

To achieve this, the EODH Data Product QA Process adheres to the ESA/NASA EO Product QA Framework [RD-1]. This framework defines requirements for a variety of aspects of Data Product quality – awarding grades of *Basic*, *Good*, *Excellent*, or *Ideal*. Such assessments are reported with a maturity matrix – this is presented in Section 0 The rest of the document provides the underpinning evidence justifying the awarded grades. This assessment has been completed by the EODH QA Service on behalf of the Data Product provider.

The result of this evaluation, referred to as a *QA Check Result*, will be added as an annotation to the EODH Catalogue for user interaction.

1.2 Changes Log

Issue	Date	Description
0.1	14/10/2025	Version for internal review
1.0	22/02/2026	Draft for pre-publication review.

1.3 Applicable Documents

[AD-1] EODH QA Service, ‘Data Product Quality Assurance Process’, 2025.

1.4 Reference Documents

[RD-1] Hunt et al., ‘Earth Observation Mission Quality Assessment Framework’ v2.2, EDAP Project Document, 2021. [Available Online](#).

[RD-2] Airbus Pléiades Neo User Guide V4. (Date of issue: August 2024)

[RD-3] Wilkinson, M., Dumontier, M., Aalbersberg, I. et al. The FAIR Guiding Principles for scientific data management and stewardship. Sci Data 3, 160018 (2016); <https://doi.org/10.1038/sdata.2016.18>

1.5 Acronyms & Abbreviations

EDAP	Earthnet Data Assessment Project
EO	Earth Observation
EODH	Earth Observation Data Hub
ESA	European Space Agency
NASA	National Aeronautics and Space Administration
NCEO	National Centre for Earth Observation
NPL	National Physical Laboratory
QA	Quality Assurance

2 Review Summary – Product Maturity Matrix

The *Data Product Quality Processes Review* is aimed at demonstrating that a given Data Product is developed and generated following processes that adhere to best practices.

To achieve this, the EODH *Data Product QA Process* adheres to the ESA/NASA EO Product QA Framework [RD-1]. Such assessments are reported in a maturity matrix style, where each box represents an area of assessment for which underpinning requirements are defined – the box colour represents the achieved grade, *Basic*, *Good*, *Excellent*, or *Ideal*. Additionally:

- *Not Assessed* means this aspect of data quality was not assessed at this time.
- *Not Assessable* means the assessment of this aspect of data quality was not possible due to a lack of available evidence.
- The padlock symbol means that the supporting evidence reviewed is not publicly available.

The *Data Product Quality Processes Review* for Airbus Pléiades Neo is presented in Figure 1. The rest of the document provides the underpinning evidence justifying the awarded grades, as follows:

- Section 3 provides the evidence for the Product Information column.
- Section 4 provides the evidence for the Metrology column.
- Section 5 provides the evidence for the Product Generation column.

Data Product Quality Processes Review			Key
Product Information	Metrology	Product Generation	
Product Details	Sensor Calibration & Characterisation	Calibration Algorithm	Not Assessed
Availability & Accessibility	Geometric Calibration & Characterisation	Geometric Processing	Not Assessable
Product Format, Flags & Metadata	Metrological Traceability Documentation		Basic
User Documentation	Uncertainty Characterisation		Good
	Ancillary Data		Excellent
			Ideal
			Not Public

Figure 1 - Data Product Quality Processes Review Maturity Matrix for Airbus Pléiades Neo

3 Product Information

Product Details	
Grade: Excellent	
Justification	Comprehensive information supplied for almost all areas in the Airbus Pléiades Neo User Guide – except no information regarding radiometric quality. In addition, further information, such as Product Locutor and Conditions for access and use, are available upon request through Data Supplier's website portal [RD-2].
Product Name	Pleiades Neo Basic MS ORT
Sensor Name	Korsch telescope, TDI instrument
Sensor Type	Optical – Multispectral and Panchromatic (6 bands + PAN)
Mission Type	Constellation – 2 satellites
Mission Orbit	Sun-synchronous, 10:30 a.m. descending node, 26 day cycle, 620 km altitude
Product Version Number	Included in product metadata
Product ID	The product naming is according to the following convention: <NUM_PRODUCT>_<MISSION><MISSION_INDEX>_<SPECTRAL_PROCESSING> Example: Dirlmage_ID = 01_PNEO3_PMS thus the image directory name is IMG_01_PNEO3_PMS Where: <NUM_PRODUCT> = Incremental product index within the Volume. Format is two digits from 01 to 99 (example 01) <MISSION><MISSION_INDEX> = Mission name (PNEO) and satellite number (3 or 4) (example PNEO3) <SPECTRAL_PROCESSING> = {PAN, MS, MS-FS, PMS, PMS-N, PMS-X, PMS-FS}
Processing level of product	Basic (digital numbers (DN) quantifying the energy recorded by the detector and corrected relatively to the other detectors to avoid non-uniformity noise), Projected (georeferenced without correction from acquisition and terrain off-nadir effects)
Measured Quantity Name	Digital Numbers (DN)
Measured Quantity Units	Counts
Stated Measurement Quality	Radiometric Quality: Not stated. Geometric Quality: Location accuracy: 3.5m CE90 @ Nadir
Spatial Resolution	Multispectral: 1.2m GSD Panchromatic: 0.3m GSD Swath width @ nadir: 14 km
Spatial Coverage	Global Period: 97.2 minutes Inclination: 97.9°

Temporal Resolution	<i>Daily</i>
Temporal Coverage	<i>10 years for each satellite, Pléiades Neo 3 & Pléiades Neo 4 – both launched in 2021.</i>
Point of Contact	<i>Contact customer service for any request: intelligence-customer_care@fr.airbus.com</i>
Product locator (DOI/URL)	<i>Products available on request via: https://space-solutions.airbus.com/contact-us/order-imagery/</i>
Conditions for access and use	<i>Depending on the service: https://space-solutions.airbus.com/legal/licences/</i>
Limitations on public access	<i>See above licences.</i>
Product Abstract	<i>From Airbus Pléiades Neo User Guide:</i> <i>Pléiades Neo is a constellation of two identical, very high-resolution optical satellites launched in 2021 on board Arianespace's Vega rockets. With a 10-year expected lifetime, they will ensure the provision of 30cm resolution imagery until at least 2031, and consequently the continuity of Airbus Intelligence's high standard geo-information services.</i>

Availability & Accessibility	
Grade: Basic	
Justification	<i>The product information, such as Product Locator and Conditions for access and use (included in product metadata, etc.) provided meets only some of the FAIR principles.</i> <i>Findable, Accessible, Interoperable and Reusable (FAIR) Principles – see [RD-4].</i>
Compliant with FAIR principles	No
Data Management Plan	Description of data management plan is limited / not available.
Availability Status	Products available on request via https://space-solutions.airbus.com/contact-us/order-imagery/

Product Format, Flags & Metadata	
Grade: Excellent	
Justification	<i>DIMAP is used which is a documented standard file format. The products also include a good set of documented metadata and data flags.</i>
Product File Format	<i>Users can select which product/image format for the Pléiades Neo imagery from:</i> <i>DIMAP – Regular and optimized JPEG 2000</i> <i>DIMAP – GeoTIFF</i> <i>NITF – Regular JPEG 2000</i> <i>NITF – GeoTIFF</i> <i>The structure for each of the formats are well-defined in Appendix A of [RD-2].</i>
Metadata Conventions	DIMAP V2 format
Analysis Ready Data?	N/A for L1 data

User Documentation		
Grade: Good		
Justification	There is a relatively recent user guide from 2024 [RD-2], but no publicly available ATBD.	
Document	Reference	QA4ECV Compliant
Product User Guide	Airbus Pléiades Neo User Guide [RD-2]	No
ATBD	Not publicly available.	N/A

4 Metrology

Sensor Calibration & Characterisation	
Grade: Basic	
Justification	<i>There is no publicly available formal sensor calibration and characterisation documentation (for either pre-flight or post-launch).</i> <i>Some high-level information available in the user guide [RD-2] states that the radiometric corrections/enhancements include:</i> <i>Inter-detector equalisation: correction of differences in sensitivity between the detectors</i> <i>Panchromatic band restored and denoised</i> <i>More detailed information on the methods used to evaluate these corrections is not publicly available.</i>
References	<i>RD-2</i>

Geometric Calibration & Characterisation	
Grade: Basic	
Justification	<i>There is no publicly available formal sensor calibration and characterisation documentation (for either pre-flight or post-launch).</i> <i>Some high-level information available in the user guide [RD-2] states that the geometric processing includes:</i> <i>The combination of all sub-swaths across the field of view (14km nadir condition): synthesized in a virtual focal plane represented by a single linear array for all spectral bands</i> <i>Correction of instrumental and optical distortions: viewing angles adjusted to the single linear array model</i> <i>Co-registration of all spectral bands: Multispectral and Panchromatic</i> <i>Attitudes and ephemeris data are refined at ground</i> <i>Adjustment of the time stamp sampling (along scan line)</i> <i>Attitude filtering over time of acquisition</i> <i>Consistent alignment of the physical location model ancillary data and RPC analytic model data</i> <i>More detailed information on the methods used to evaluate these corrections is not publicly available.</i>
References	<i>RD-2</i>

Metrological Traceability Documentation	
Grade: Not Assessable	
Justification	<i>There are no metrology documentation or traceability chain / uncertainty tree diagrams publicly available.</i>
References	N/A

Uncertainty Characterisation	
Grade: Not Assessable	
Justification	<i>There is no detailed information describing regarding the methods used to characterise the uncertainties publicly available.</i>
References	N/A

Ancillary Data	
Grade: Good	
Justification	<i>An overview of per product ancillary data (e.g. viewing and solar angles, longitude, latitude) is available in the user guide, Appendices A & D [RD-2], explaining its location within product metadata.</i> <i>However, there is no detailed information regarding uncertainty characterisation for any of the ancillary data.</i> <i>Specifically, DEM data used in ortho-rectification is specified at collection level, and not per product. In the user guide [RD-2], there is an overview of the geometric details used in the ORTHO product:</i> <i>Geographic projections: WGS84 – latitude/longitude sampling according to DTED2 zoning standards (please refer to Appendix G: DTED for more details)</i> <i>Mapping projections: Most of the projections registered by EPSG, in meters (please refer to A.7.2 for more details)</i> <i>GCP: Space Reference Points (SRP), specific user GCP may be supplied for tailored products</i> <i>DEM: WorldDEM4Ortho DSM/DTM, Reference3D DEM layer (DTED2), SRTM (DTED1), GLOBE (DTED0)</i> <i>An RPC XML file (containing the coefficients and Normalisation parameters for the Rational Polynomial Coefficients (also called Rapid Positional Capability, Rational Function Model) geometric analytic model) is also provided which allows users to carry out geometric processing with software that supports RPC models.</i>

References	• <i>RD-2</i>
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5 Product Generation

Calibration Algorithm	
Grade: Basic	
Justification	Information on the radiometric and atmospheric corrections given in Appendix E.4 of the User Guide with description of input parameters and where to access them in the XML metadata file. However, not thorough enough to deem “fit for purpose” in terms of the mission’s stated performance.
References	RD-2

Geometric Processing	
Grade: Good	
Justification	Information on the geometric processing given in Appendix D of the User Guide with description of input parameters and where to access them in the XML metadata file. The described algorithm is reasonable for the mission’s stated performance, and the calibration quality is considered sufficient.
References	RD-2